

Transformasi Intensitas

INFORMATIKA UNS – PENGOLAHAN CITRA DIGITAL GANJIL 2026/2027

Spatial Domain dan Transformasi Intensitas

Spatial Domain dan Transformasi Intensitas

Apa itu *spatial domain*?

- The term spatial domain refers to the image plane itself, and image processing methods in this category are based on direct manipulation of pixels in an image.
- Teknik domain spasial beroperasi langsung pada piksel suatu citra.

Spatial Domain dan Transformasi Intensitas

Dasar-dasar Transformasi Intensitas & Spatial Filtering (1)

- Spatial domain pada bahasan kali ini mengacu pada:

$$g(x, y) = T[f(x, y)] \quad (3-1)$$

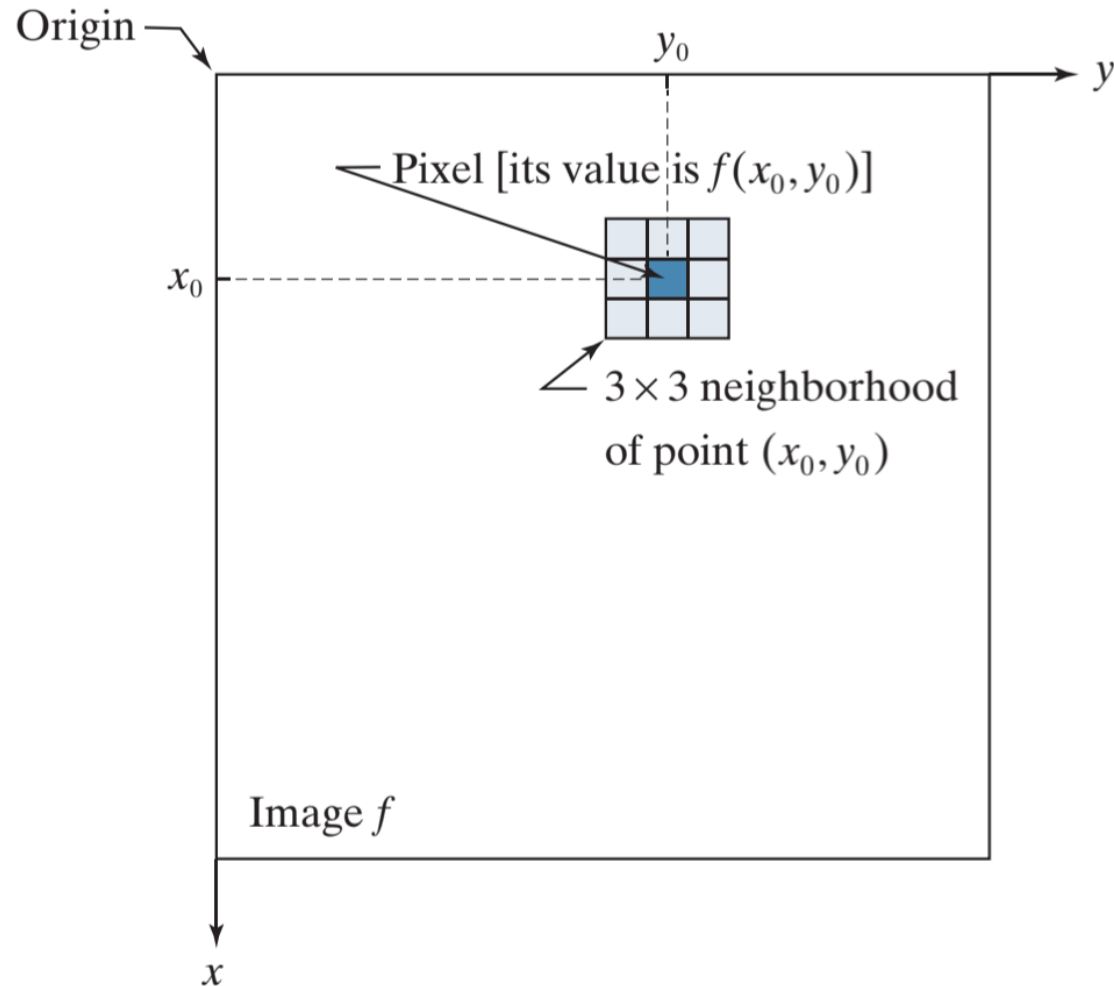
- Dengan:
 - $f(x, y)$ is an input image
 - $g(x, y)$ is an output image
 - T is an operator on f defined over a neighborhood of point (x, y) .
- Jadi, T bisa beroperasi pada satu pixel, sekelompok pixel yang saling bertetangga, atau seluruh piksel pada citra.

Spatial Domain dan Transformasi Intensitas

Dasar-dasar Transformasi Intensitas & Spatial Filtering (2)

FIGURE 3.1

A 3×3 neighborhood about a point (x_0, y_0) in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image. Recall from Chapter 2 that the value of a pixel at location (x_0, y_0) is $f(x_0, y_0)$, the value of the image at that location.



Spatial Domain dan Transformasi Intensitas

Dasar-dasar Transformasi Intensitas & Spatial Filtering (3)

- The smallest possible neighborhood is of size 1×1 . In this case, g depends only on the value of f at a single point (x, y) and T in Eq. (3-1) becomes an intensity (also called a gray-level, or mapping) transformation function of the form:

$$s = T(r) \quad (3-2)$$

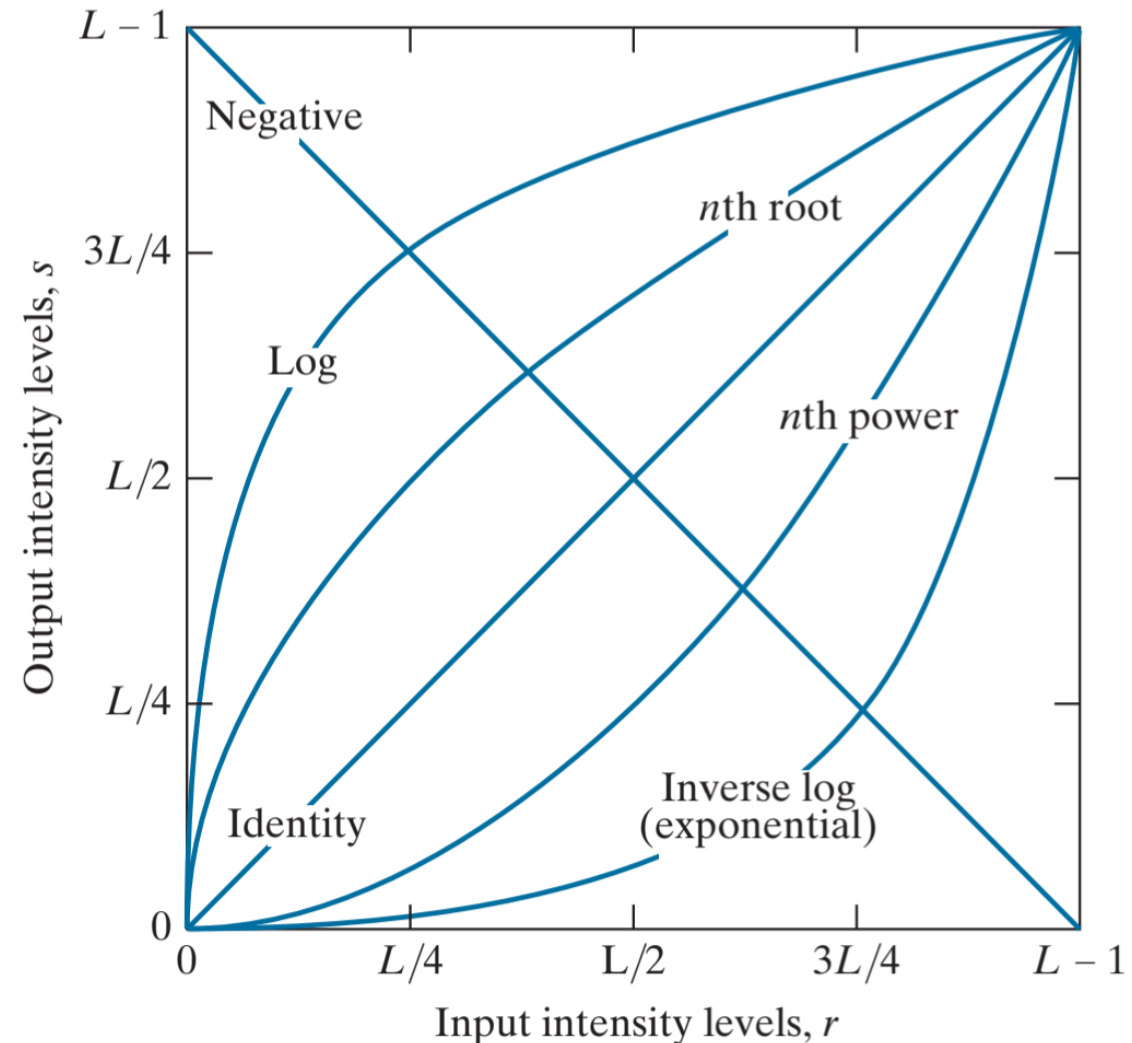
- For simplicity in notation, we use s and r to denote, respectively, the intensity of g and f at any point (x, y) .
- Disini, T hanya beroperasi pada piksel tunggal.

Beberapa Dasar Transformasi Intensitas

Beberapa Dasar Transformasi Intensitas Citra Negatif (Image Negative) (1)

FIGURE 3.3

Some basic intensity transformation functions. Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



Beberapa Dasar Transformasi Intensitas

Citra Negatif (Image Negative) (2)

- Bentuk negative dari suatu citra dengan level intensitas pada rentang $[0, L - 1]$ dapat diperoleh menggunakan fungsi transformasi negatif seperti dalam Fig. 3.3, yang dapat diformulasikan:

$$s = L - 1 - r \quad (3-3)$$

Beberapa Dasar Transformasi Intensitas

Citra Negatif (Image Negative) (3)

- Membalikkan tingkat intensitas citra digital dengan cara ini menghasilkan hasil yang setara dengan fotografi negatif.
- Jenis pemrosesan ini misalnya digunakan untuk meningkatkan detail putih atau abu-abu yang berada di daerah gelap suatu gambar, terutama ketika area hitam berukuran dominan.

Beberapa Dasar Transformasi Intensitas

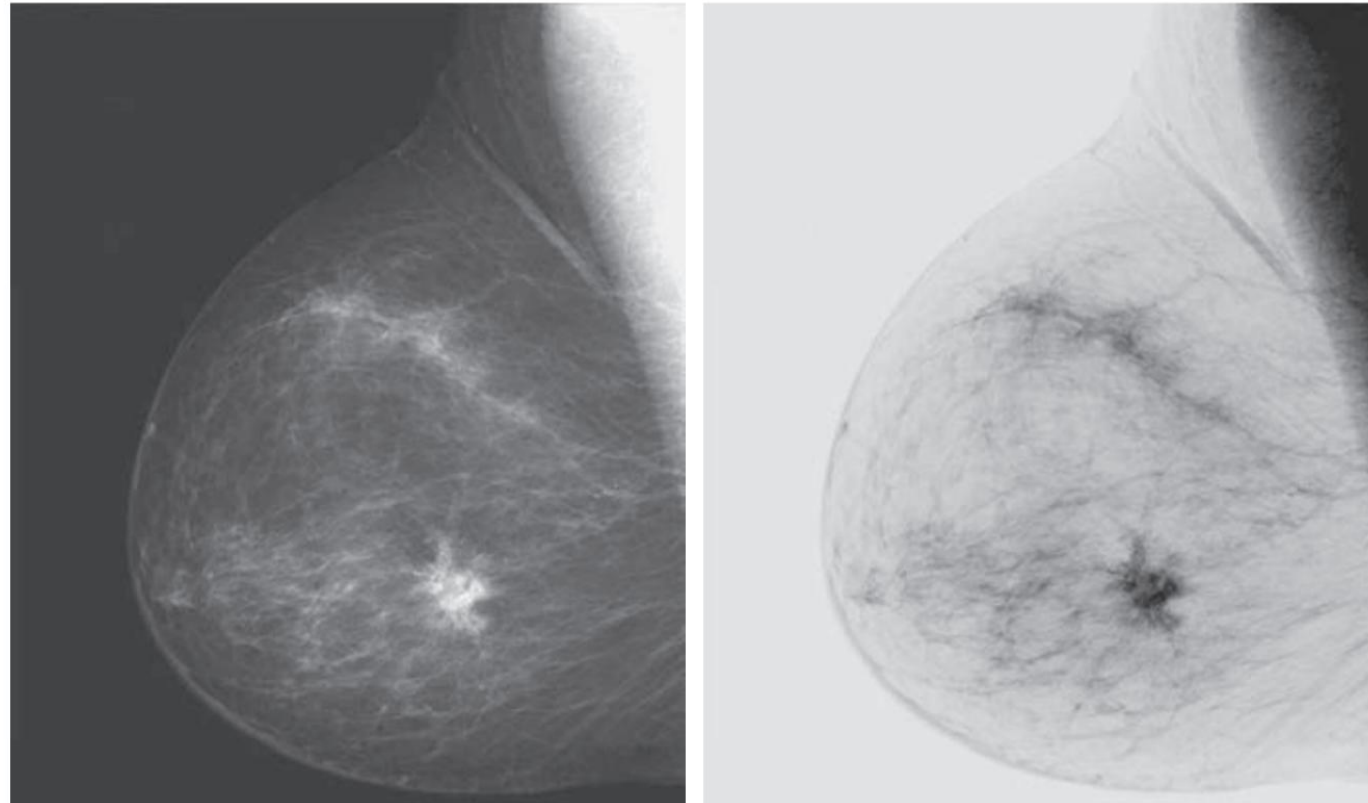
Citra Negatif (Image Negative) (4)

- Figure 3.4 shows an example. The original image is a digital mammogram showing a small lesion. Despite the fact that the visual content is the same in both images, some viewers find it easier to analyze the fine details of the breast tissue using the negative image.

a b

FIGURE 3.4

(a) A digital mammogram.
(b) Negative image obtained using Eq. (3-3).
(Image (a) Courtesy of General Electric Medical Systems.)



Beberapa Dasar Transformasi Intensitas Citra Negatif (Image Negative) (5)

- [LATIHAN]

- Diberikan suatu citra grayscale 8-bit dalam bentuk matriks berikut:

$$A = \begin{bmatrix} 55 & 165 & 110 \\ 110 & 165 & 55 \\ 220 & 55 & 220 \end{bmatrix}$$

- Tentukan matriks hasil citra negatifnya!

Beberapa Dasar Transformasi Intensitas

Transformasi Logaritma (1)

- Bentuk transformasi logaritma seperti dalam Fig. 3.3:

$$s = c \log(1 + r) \quad (3-4)$$

- Dengan c adalah konstanta dan $r \geq 0$.
- Transformasi ini digunakan untuk memperluas nilai piksel gelap dalam suatu citra, sambil mengompresi nilai pada level yang lebih tinggi.
- Hal yang sebaliknya berlaku pada transformasi invers logaritma (eksponensial):

$$s = e^{(r/c)} - 1$$

Beberapa Dasar Transformasi Intensitas

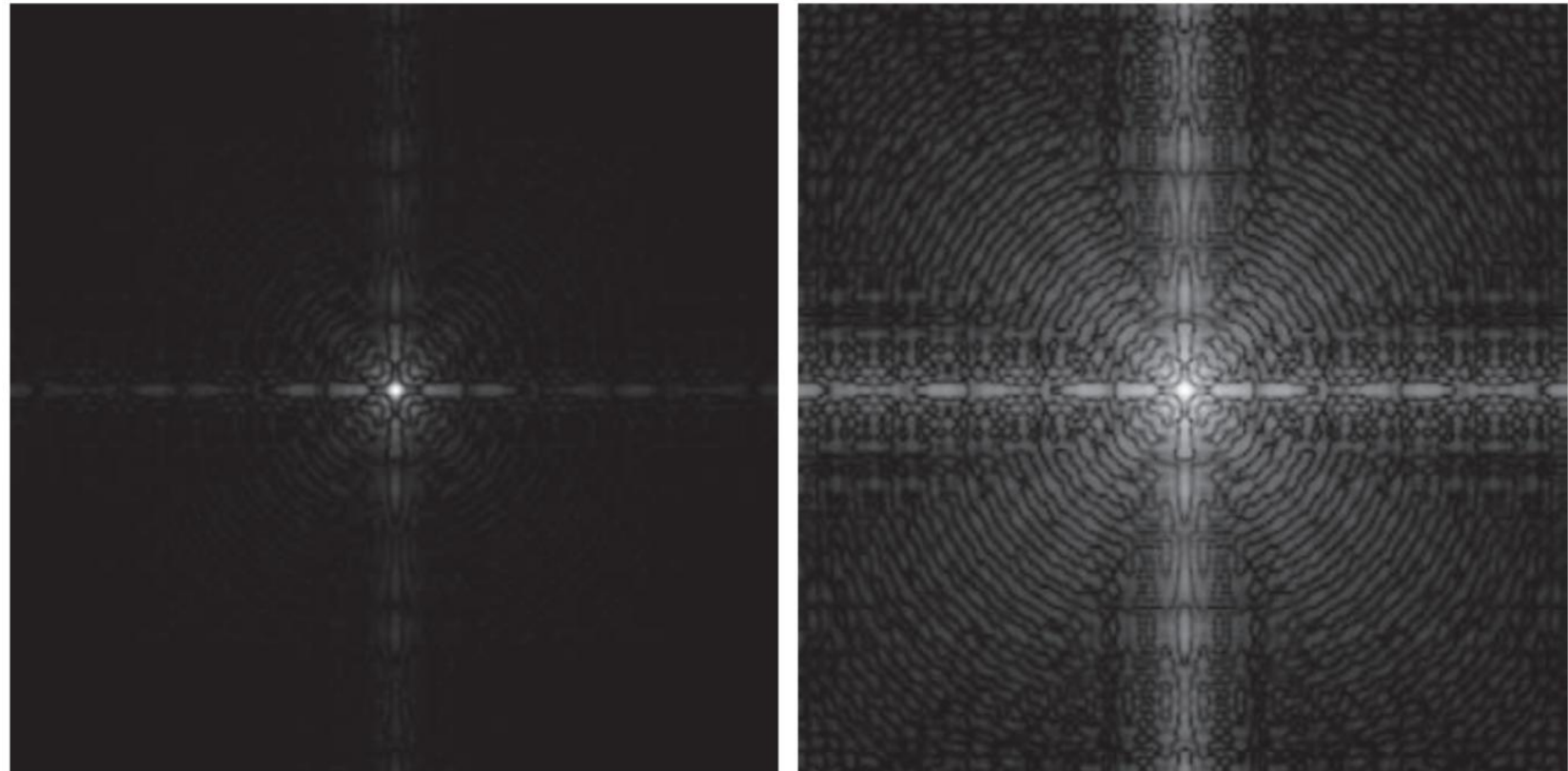
Transformasi Logaritma (2)

- The log function has the important characteristic that it compresses the dynamic range of pixel values. An example in which pixel values have a large dynamic range is the Fourier spectrum.

a b

FIGURE 3.5

(a) Fourier spectrum displayed as a grayscale image.
(b) Result of applying the log transformation in Eq. (3-4) with $c = 1$. Both images are scaled to the range $[0, 255]$.



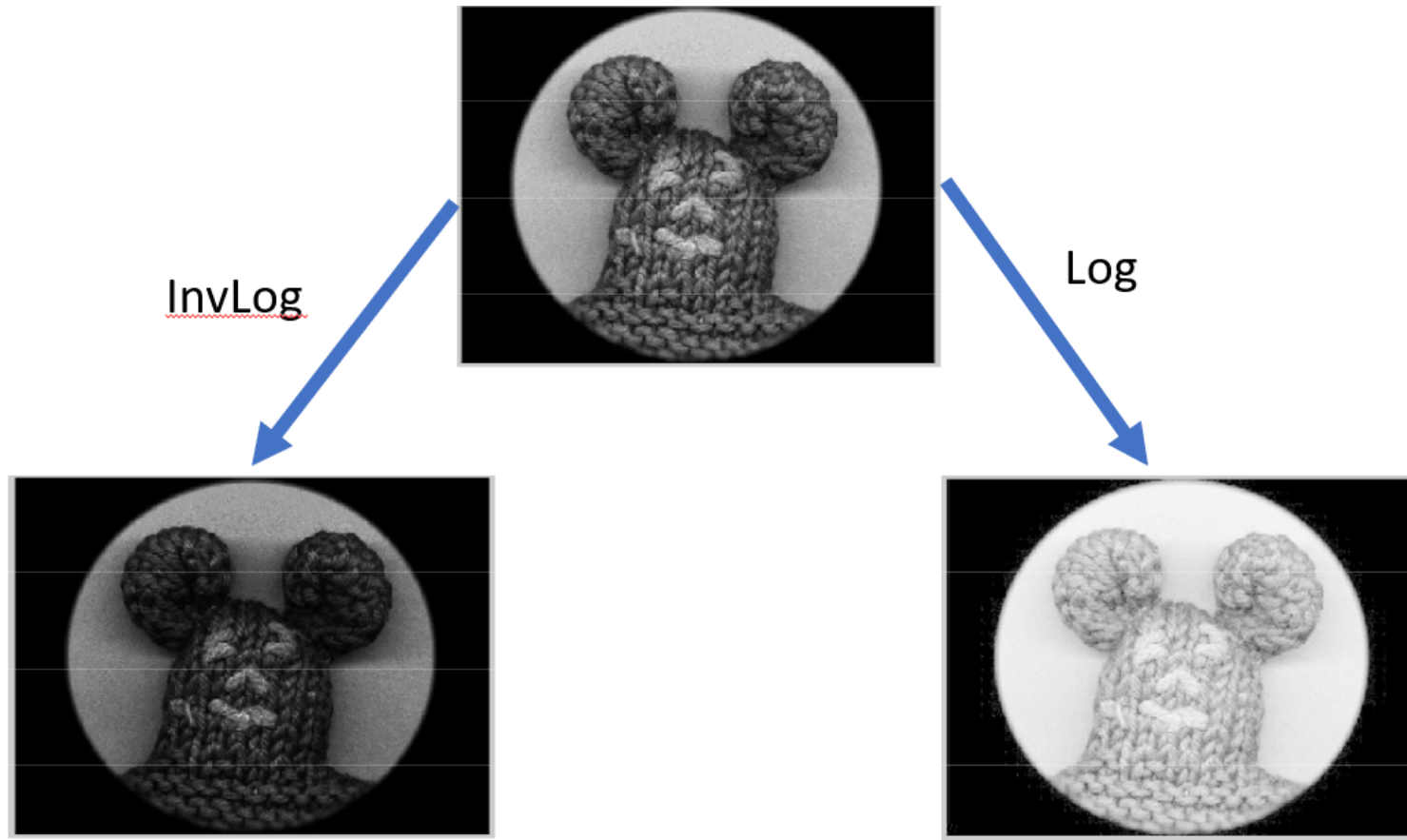
Beberapa Dasar Transformasi Intensitas

Transformasi Logaritma (3)

- Figure 3.5(a) shows a Fourier spectrum with values in the range **0 to 1.5×10^6** . When these values are scaled linearly for display in an 8-bit system, the brightest pixels dominate the display, at the expense of lower (and just as important) values of the spectrum.
- The effect of this dominance is illustrated vividly by the relatively small area of the image in Fig. 3.5(a) that is not perceived as black. If, instead of displaying the values in this manner, we first apply Eq. (3-4) (with $c = 1$ in this case) to the spectrum values, then the range of values of the result becomes **0 to 6.2**.
- Transforming values in this way enables a greater range of intensities to be shown on the display.

Beberapa Dasar Transformasi Intensitas

Transformasi Logaritma (4)



Beberapa Dasar Transformasi Intensitas

Power-law (Gamma) Transformation (1)

- Power-law (Gamma) Transformation diformulasikan sebagai:

$$s = cr^{\gamma} \quad (3-5)$$

- Dengan c dan γ adalah konstanta positif.
- Banyak device yang digunakan untuk image capture, printing, dan display mendukung penggunaan power-law.
- Proses yang digunakan untuk mengoreksi power-law disebut *gamma correction* atau *gamma encoding*.

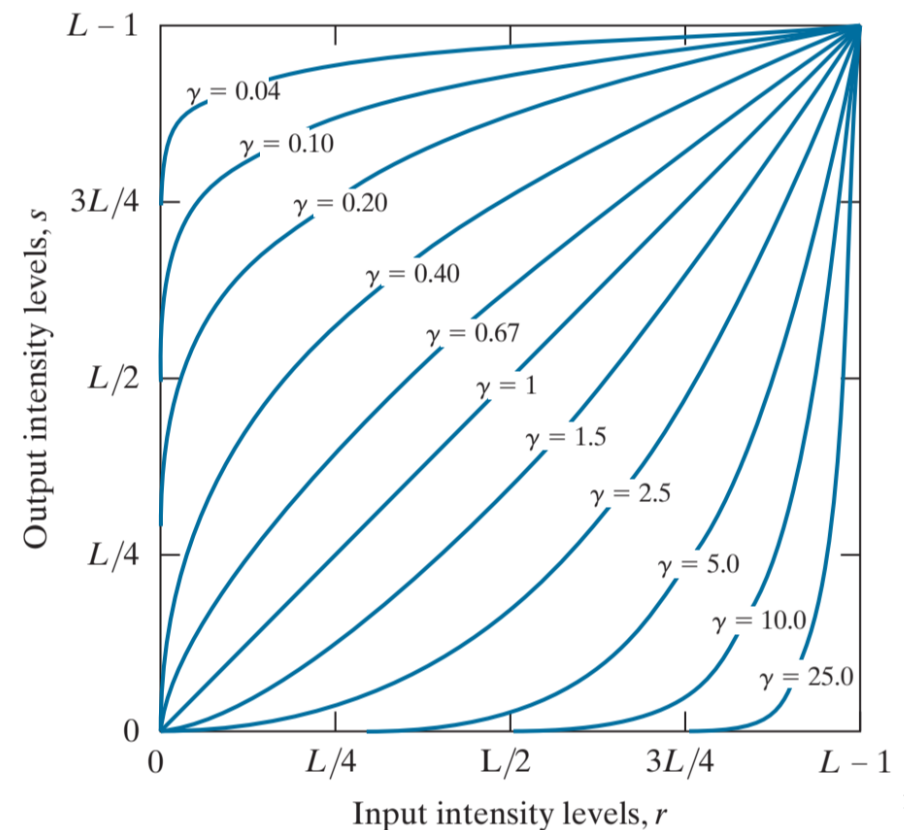
Beberapa Dasar Transformasi Intensitas

Power-law (Gamma) Transformation (2)

- Note also in Fig. 3.6 that a family of transformations can be obtained simply by varying γ . Curves generated with values of $\gamma > 1$ have exactly the opposite effect as those generated with values of $\gamma < 1$. When $c = \gamma = 1$ Eq. (3-5) reduces to the identity transformation.

FIGURE 3.6

Plots of the gamma equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



Beberapa Dasar Transformasi Intensitas Power-law (Gamma) Transformation (3)

a	b
c	d

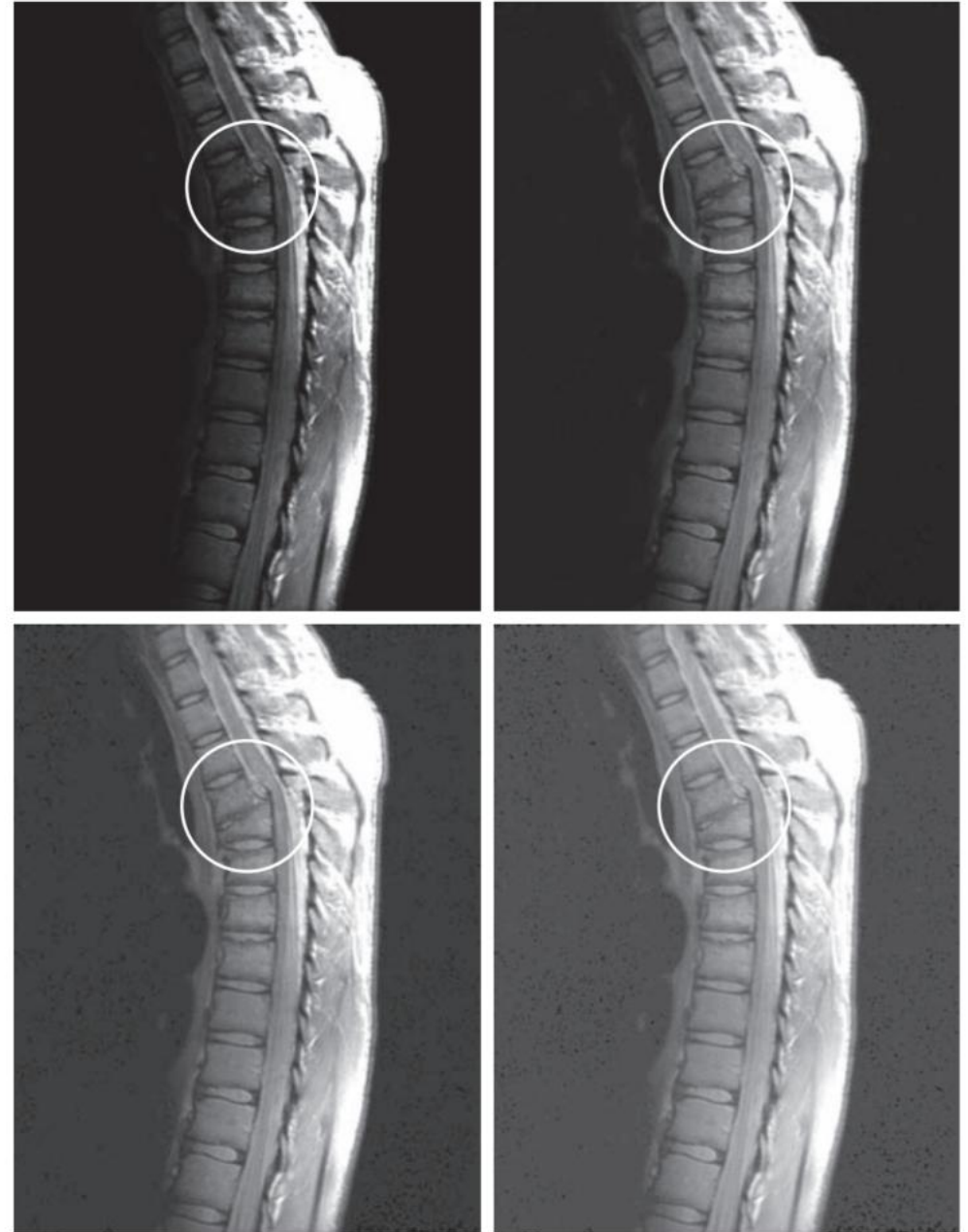
FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5)

with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



Beberapa Dasar Transformasi Intensitas

Power-law (Gamma) Transformation (4)

a	b
c	d

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of applying the transformation in Eq. (3-5) with $\gamma = 3.0, 4.0$, and 5.0 , respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)



Beberapa Dasar Transformasi Intensitas

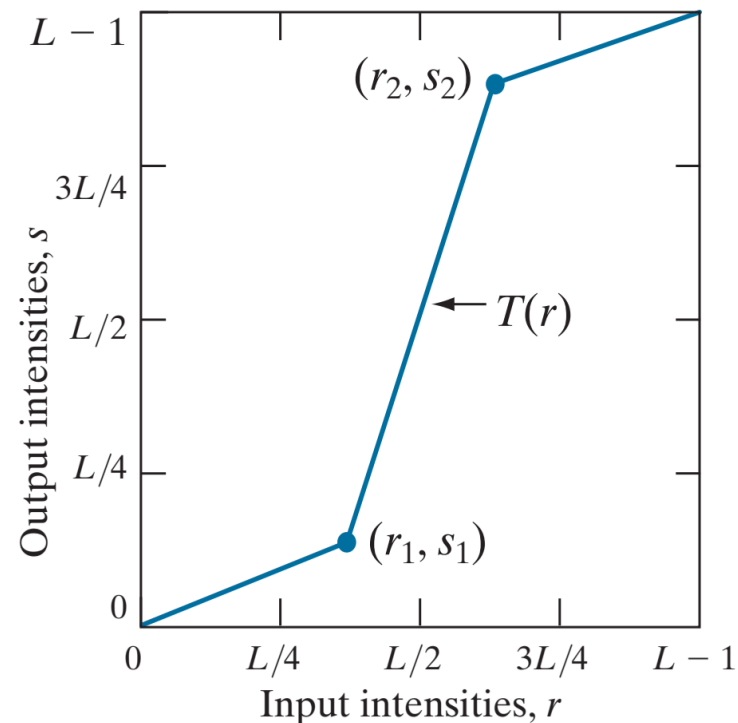
Piecewise Linear Transformation Functions

- Diantara yang termasuk dalam piecewise linear transformation functions (fungsi transformasi linier sepotong-sepotong):
 1. Contrast Stretching
 2. Intensity-Level Slicing

Beberapa Dasar Transformasi Intensitas

Contrast Stretching (1)

- Contrast Stretching (peregangan kontras): meningkatkan rentang nilai keabuan pada citra dengan kontras rendah (rentang dari r_1 sampai r_2 pada citra dengan nilai keabuan 0 sampai $L - 1$).



Beberapa Dasar Transformasi Intensitas

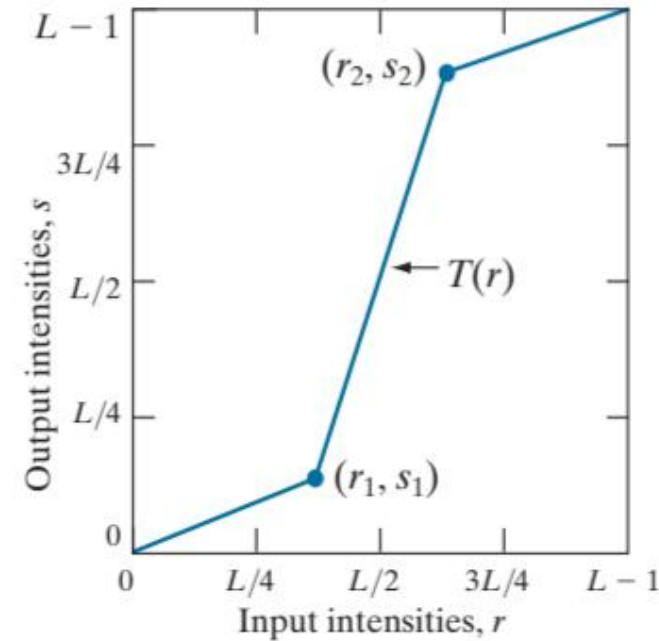
Contrast Stretching (2)

a	b
c	d

FIGURE 3.10

Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

- Figure 3.10(c) shows the result of contrast stretching, obtained by setting:
- $(r_1, s_1) = (r_{\min}, 0)$ and $(r_2, s_2) = (r_{\max}, L-1)$
- Where $r_{\min}$ and $r_{\max}$ denote the minimum and maximum intensity levels in the input image, respectively.



Beberapa Dasar Transformasi Intensitas

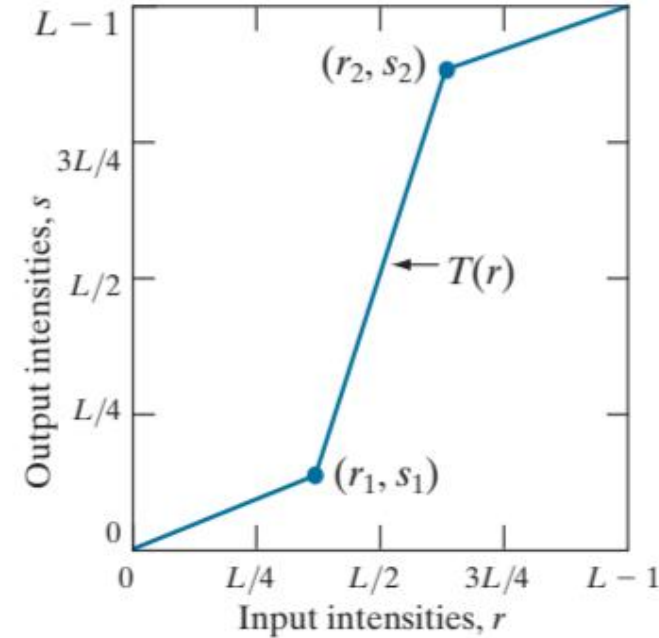
Contrast Stretching (3)

a	b
c	d

FIGURE 3.10

Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

- Fig. 3.10(d) shows the result of using the thresholding function, with:
- $(r_1, s_1) = (m, 0)$ and $(r_2, s_2) = (m, L-1)$
- Where m is the mean intensity level in the image.



Beberapa Dasar Transformasi Intensitas

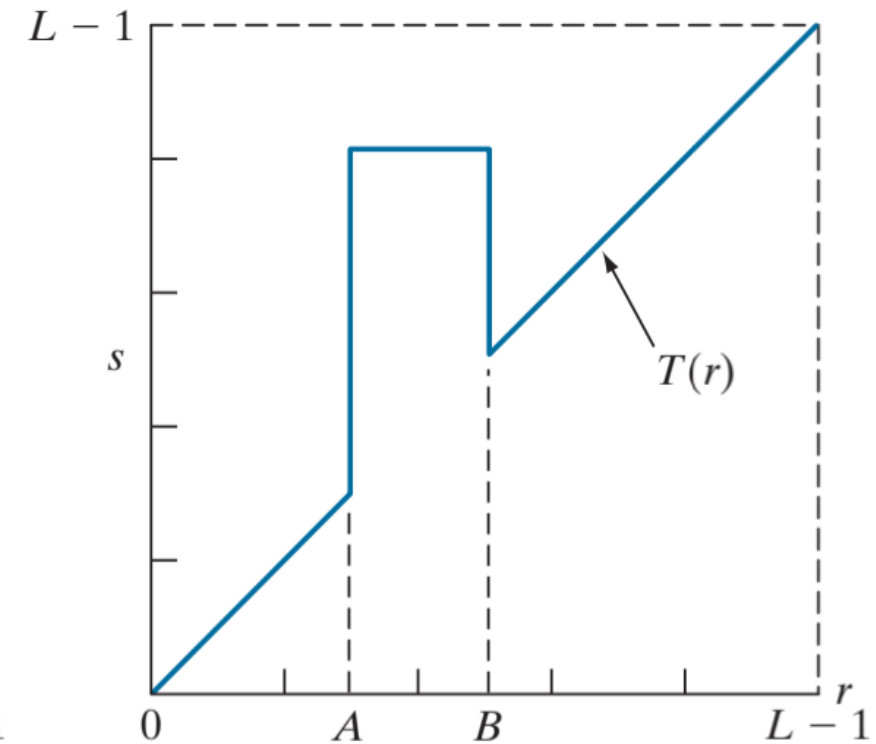
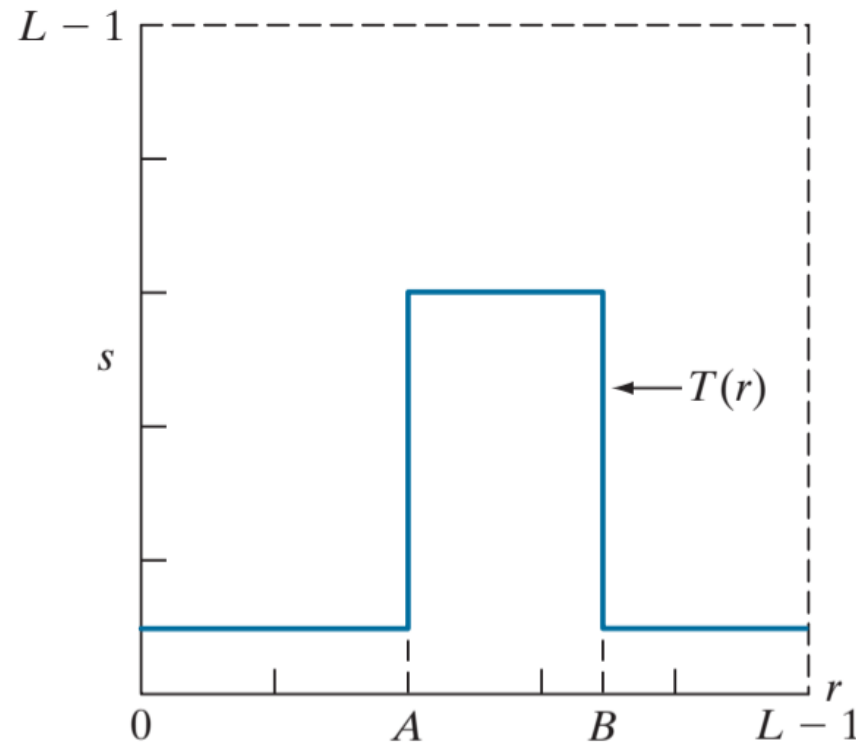
Intensity-Level Slicing (1)

- Menonjolkan keabuan pada rentang tertentu pada citra

a b

FIGURE 3.11

(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level.
(b) This function highlights range $[A, B]$ and leaves other intensities unchanged.



Beberapa Dasar Transformasi Intensitas

Intensity-Level Slicing (2)



a b c

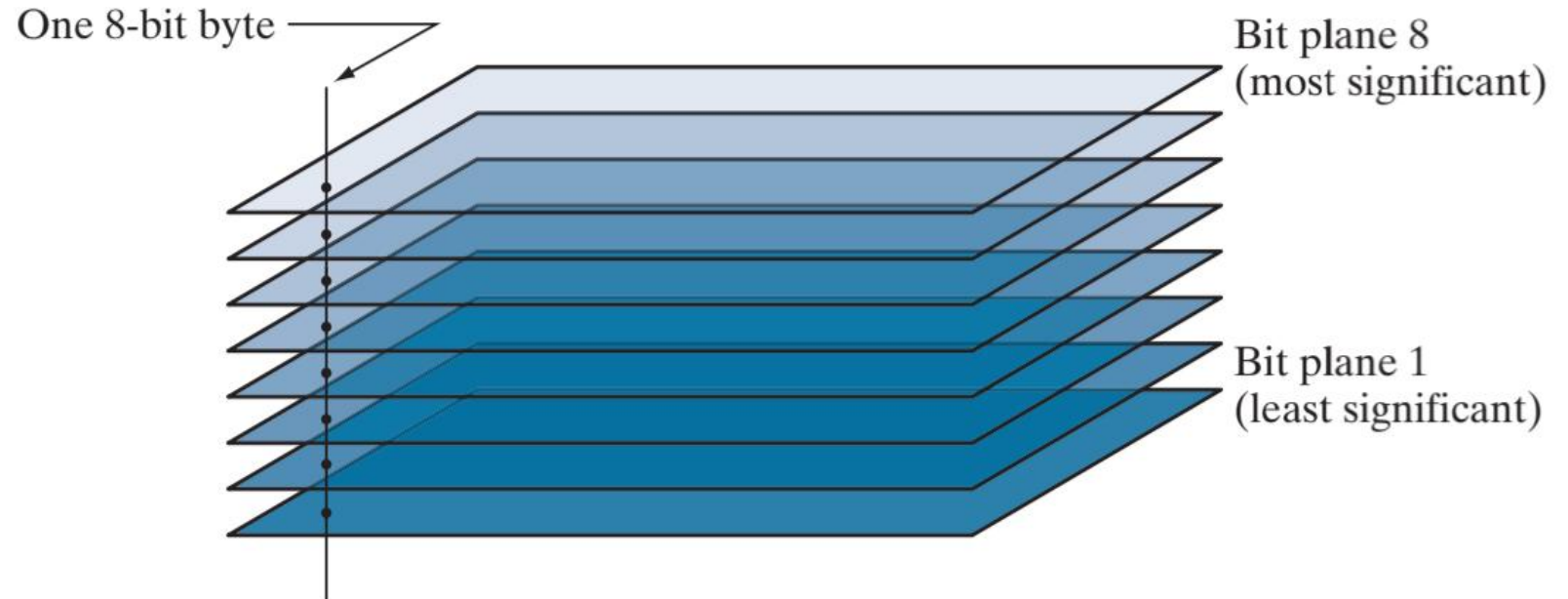
FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing (1)

- As Fig. 3.13 illustrates, an 8-bit image may be considered as being composed of eight one-bit planes, with plane 1 containing the lowest-order bit of all pixels in the image, and plane 8 all the highest-order bits.

FIGURE 3.13
Bit-planes of an
8-bit image.



Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing (2)

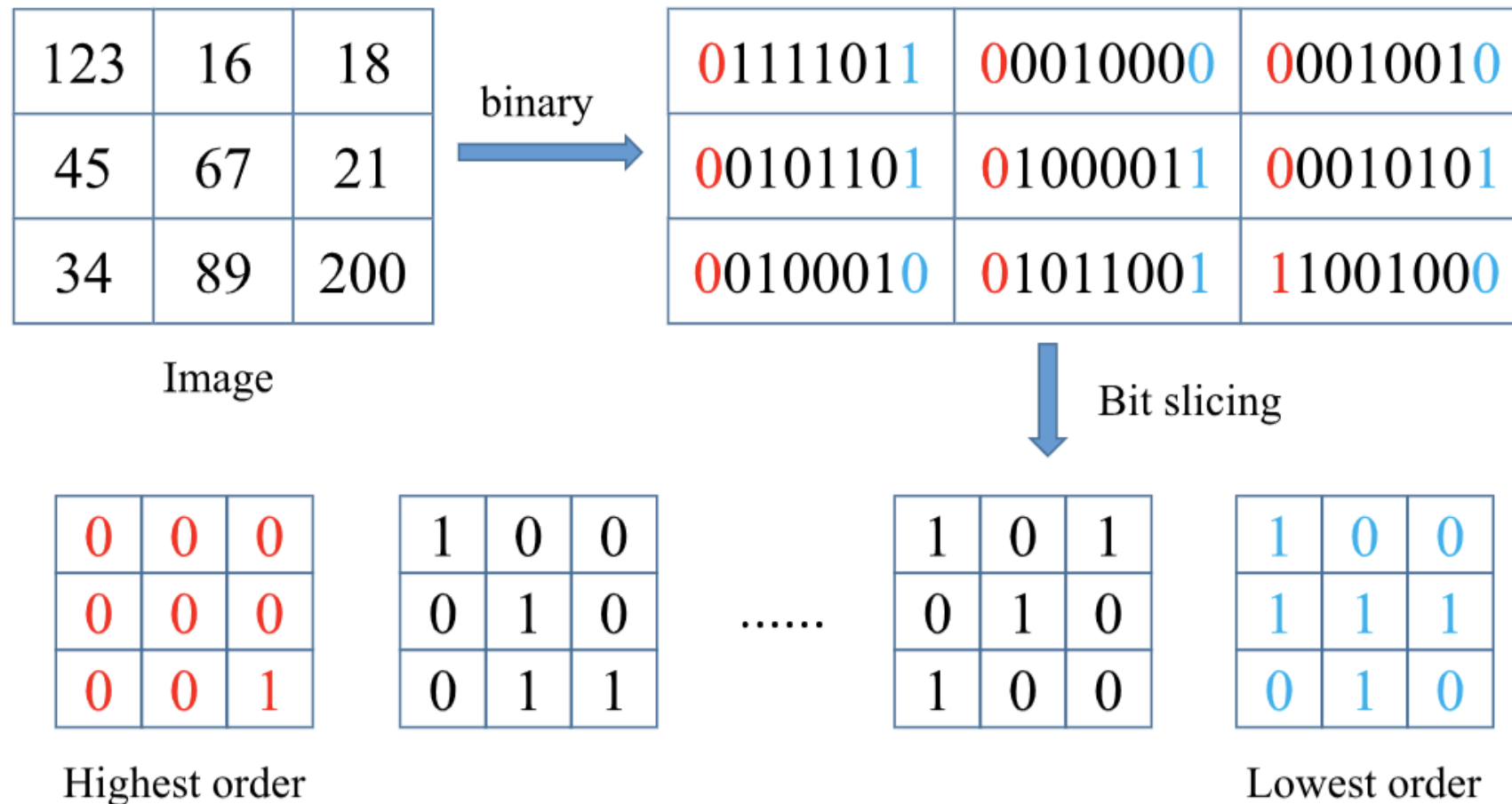
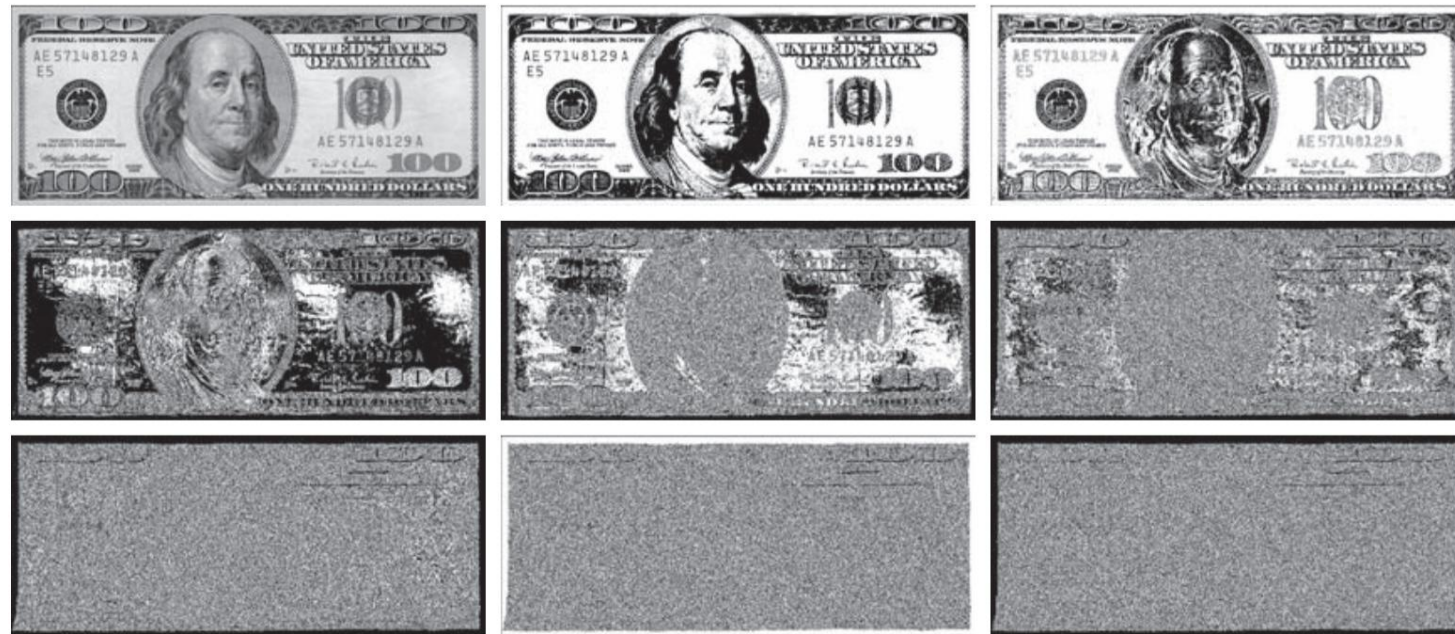


Fig. 4. Process of bit-plane slicing.

Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing (3)

- Figure 3.14(a) shows an 8-bit grayscale image and Figs. 3.14(b) through (i) are its eight one-bit planes, with Fig. 3.14(b) corresponding to the highest-order bit. Observe that the four higher-order bit planes, especially the first two, contain a significant amount of the visually-significant data.



a	b	c
d	e	f
g	h	i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing (4)

- Decomposing an image into its bit planes is useful for analyzing the relative importance of each bit in the image, a process that aids in determining the adequacy of the number of bits used to quantize the image.
- Also, this type of decomposition is useful for image compression, in which fewer than all planes are used in reconstructing an image.



a b c **FIGURE 3.15** Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.

Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing – Rekonstruksi (1)

1. *Plane* ke- n dalam piksel dikalikan dengan konstanta $2^{(n-1)}$.
2. Sebagai contoh, dari matriks berikut ini, kita akan merekonstruksi menggunakan bit *plane* 8 dan 7:

$$\begin{bmatrix} 167 & 133 & 111 \\ 144 & 140 & 135 \\ 159 & 154 & 148 \end{bmatrix}$$

Didapatkan format bit masing-masing:

10100111	10000101	01101111
10010000	10001100	10000111
10011111	10011010	10010100

Beberapa Dasar Transformasi Intensitas

Bit-Plane Slicing – Rekonstruksi (2)

3. Gabungkan bit *plane* 8 dan bit *plane* 7:

Untuk 10100111, kalikan bit *plane* ke 8 dengan $2^{(8-1)} = 128$ dan bit *plane* ke 7 dengan $2^{(7-1)} = 64$:

$$(1 \times 128) + (0 \times 64) + (1 \times 0) + (0 \times 0) + (0 \times 0) + (1 \times 0) + (1 \times 0) + (1 \times 0) = 128$$

4. Ulangi proses ini untuk semua nilai dalam matriks, dan hasil akhirnya adalah:

$$\begin{bmatrix} 128 & 128 & 64 \\ 128 & 128 & 128 \\ 128 & 128 & 128 \end{bmatrix}$$

Terima Kasih
